Todo list

The mousik package*

Celia Rubio Madrigal†

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Abstract

This is a music-related package focused on writing music fragments. It works specially well with small examples, such as excerpts for theoretical articles. It is easier than other music writing methods in LaTeX because it is based on the tikz package.

Keywords

mousik, music, music theory, music notation, music writing, score, sheet, tikz diagram, bar, clef, time signature, note, symbol, rest, dot, chord, accidental, sharp, flat, natural, key

Contents

*This document corresponds to mousik v0.1, dated 2020/07/01.

†Email: celirubio.m@gmail.com

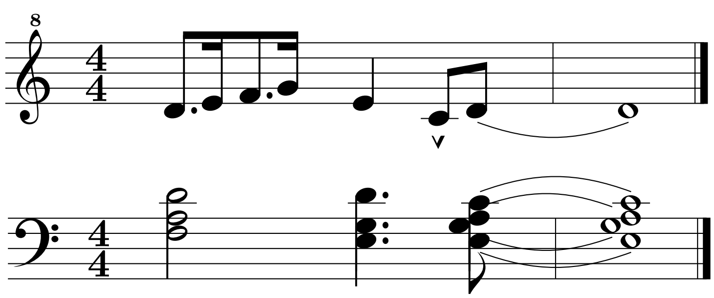
1 Introduction

There are already several ways to set music using LaTeX. As stated by the abc package, "*notably MusixTEX and Lilypond*", as well as abc per se. Some of them are too difficult to use for not too complicated music; others do not work at all.

The mousik package is perfect for short, one line music fragments. It may be used for longer scores, but there are more adaptable and lighter packages than this one. However, in spite of its compactness, the package is quite complete and versatile, but easy to learn and use.

2 The `mousik` environment

The way to draw a score is by means of the `mousik` environment. For longer examples, see section 5.



```
\begin{mousik}
   \Clef
   \TimeSig{2}{4}

   \Dots{*/1, , */3, }
   \Note{|-,=|,|-,=|}{1,2,3,4}
   \Bar

   \Note{4}{2}
   \Artic{^}{0}
   \Note{8}{0,1}
   \Bar[|.]
\end{mousik}
```

It has some options:

- *indent* indents the first line. Default is 0.

- *long* changes the maximum width of the score. Default is 100.

piano scores

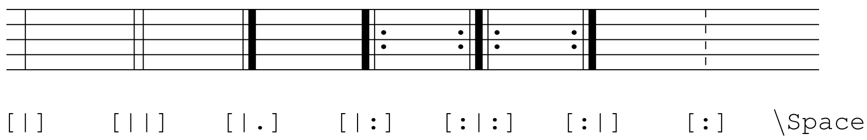
single staff

space between notes

3 Available commands

3.1 \Bar

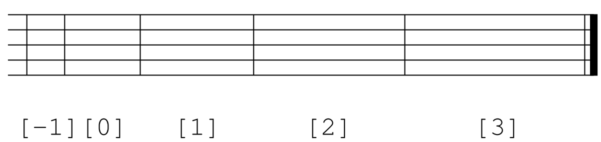
The `\Bar` command prints a bar on the score. It has an optional argument depending on the type of bar, with default `|`, the single bar (`\Bar` = `\Bar[|]`).



The only way to end a line in the score is with a bar or a space. Any other way may cause visual errors (see subsection 4.1).

3.2 \Space

The `\Space` command is not only used when no bar is needed. It may also be used to input *extra* space between symbols. The default space is 2 (which is the default space between symbols in general), but it may be any whole number (see subsection 4.2).

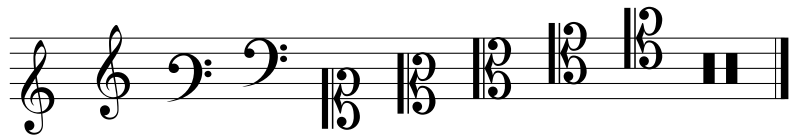


xshift yshift

3.3 \Clef

The `\Clef` command might be used at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1). It has some options:

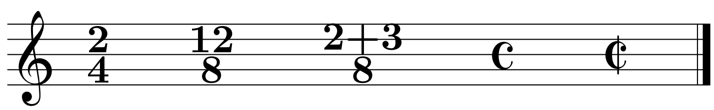
- *octave* prints an octave eight on the clef. If it is an 8, it prints it above the clef; if it is a -8, it prints it below the clef. Default is 0.
- *t* (type) is the type of clef. Default is 2 (the Treble Clef).



1 2 3 4 11 12 13 14 15 0

3.4 \TimeSig

Time signatures might be used at any moment. `\TimeSig{2}{4}` will draw a 2 by 4 time signature. Any number or symbol can be written inside the brackets. Also `\TimeSig{C}{}` will produce a *common time* symbol and `\TimeSig{C|}{}` will produce an *alla breve* symbol.



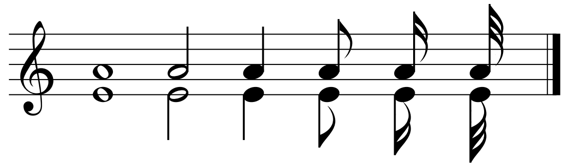
3.5 Notes and rests

3.5.1 \Note

The `\Note` command prints a note on the score. It accepts 2 compulsory arguments, as well as some options:

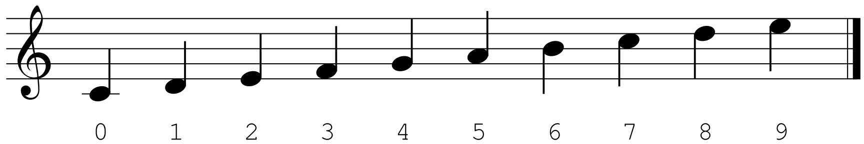
- *beam* chooses the width of the beam. Default is 1 (mm).
- *head* chooses the type of head. Default is . (round), but also x (cross) and o (hollow) are possible.
- *stem* chooses the height of the stem. Default is 0.3.
- *swap* prints the note upside down. Default is false, which means up if the height is less than 6, down otherwise.

The first compulsory argument is the duration of the note.



1 2 4 8 16 32

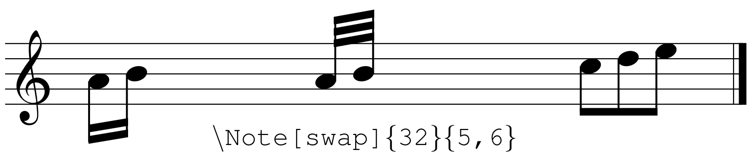
The second argument is the absolute height. For example, these are the notes from 0 to 9:



remove white

The \Note command might also print several notes tied together with a beam, if the second argument is a list of heights separated by commas. Two notes will be tied together with an oblique beam, but three or more will have a straight beam.

tuplet

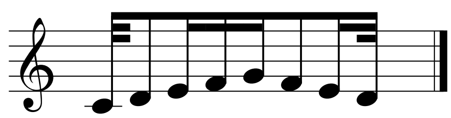


\Note[swap]{32}{5,6}

\Note[stem=0.4]{16}{5,6}   \Note{8}{7,8,9}

Notes with different durations that are tied together have a different syntax. Instead of using 8, 16 and 32 for the duration, each note has to be referenced with a -, a = or a ≡, respectively, and additional | might be needed for half the beam.

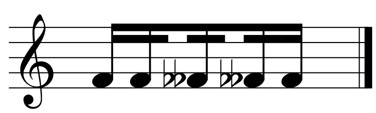
explain better



\Note{|≡,-,|=,=,=|,-,|=,≡|}{0,1,2,3,4,3,2,1}

To be able to write the ≡ as a symbol on the .tex file, I usually copy and paste it wherever I need it. Nonetheless, an alternative keyboard-friendly input that is equally valid is using /// instead of ≡.

Additionally, spaces between notes from the same group might be useful —the default space is 1. A space of 1.5 is obtained when a ! is placed on both lists. A space of 2 is obtained when a space is placed on both lists.

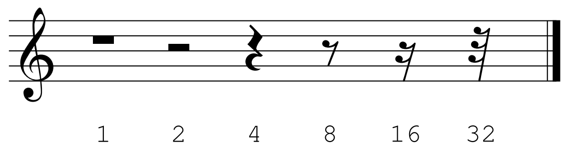


\Accs{,,!,-2/3}
\Note{|=,=|,!,-, ,=|}{3,3,!,3, ,3}
\Space[-3]\Rest{16}\Space[1]

fix beam =

3.5.2 \Rest

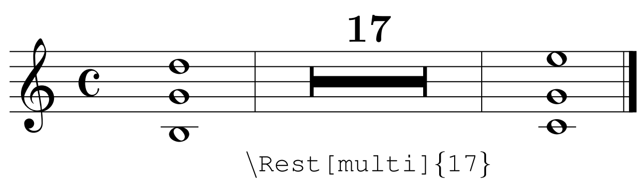
The \Rest command prints a rest. It accepts its duration as an argument: 1, 2, 4, 8, 16 or 32.



common height units

It also has some options:

- *height* moves the rest up and down. Default is 0.
- *multi* transforms the rest into a multimeasure rest —where the rest lasts an arbitrary amount of bars.

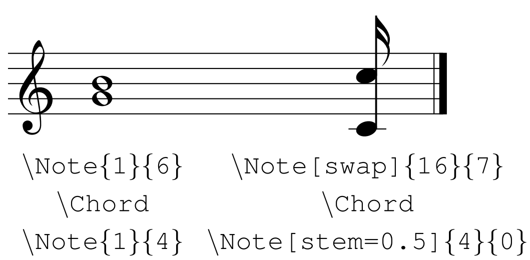


3.5.3 \Chord

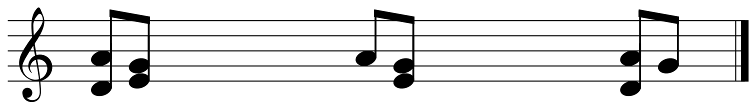
Chords may be obtained by means of the \Chord command. It accepts an optional argument: the number of notes to step backwards —mostly for the \Note command— with 1 as default.

maybe fix chord

The best way to use it is to call the \Note command with the highest notes; then call the \Chord command; then call the \Note command with the duration 4 of a quarter note, to print them on top.



For extreme uses of the command, extra \Space commands might be needed.



\Note{8}{5,4}	\Note{8}{5,4}	\Note{8}{5,4}
\Chord[2]	\Chord	\Chord[2]
\Note{4}{1,2}	\Note{4}{2}	\Note{4}{1}
		\Space[1]

maybe fix chord+space

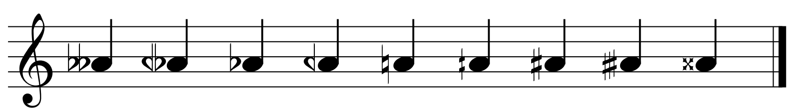
3.6 Keys and accidentals

3.6.1 \Acc

The `Acc` command prints an accidental before a note. It accepts 2 arguments.

The first argument is the type of accidental: a number ranging from -2 to 2. The flat ♭ symbol might also be called with a `b` and the sharp ♯ symbol might be called with `\#`.

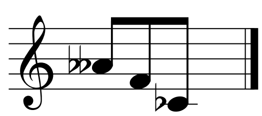
only # ?



−2	−1.5	−1	−0.5	0	0.5	1	1.5	2
		b				\#		

The second argument is the height of the symbol.

When there are several notes, the `\Accs` command might be useful. It accepts as an argument a list of pairs: type/height. Where there is no accidental, the spot shall be left empty.

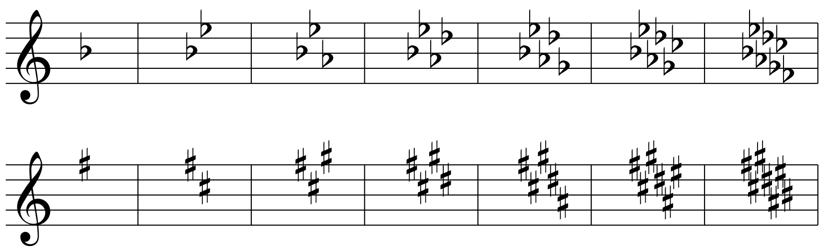


`\Accs{-2/5, ,-1/0}`
`\Note{5,3,0}`

3.6.2 \Key

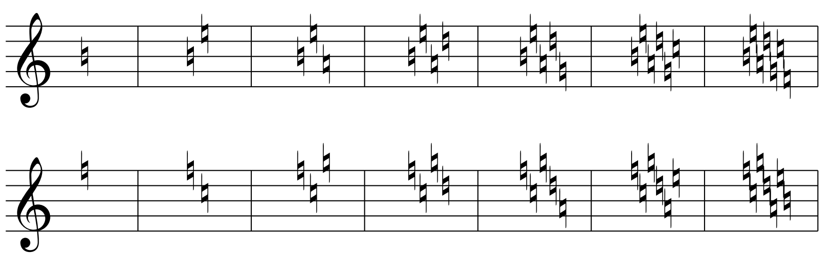
The `\Key` command prints the key at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1).

It accepts an argument: a positive whole number from 1 to 7 for sharps ♯, and a negative whole number from -1 to -7 for flats ♭.

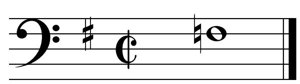


3.6.3 \NoKey

The `\NoKey` command is the same as the `\Key` command, but it prints naturals ♮ instead of sharps or flats. Mixed Keys



Be careful: the `\Key` and `\NoKey` commands are affected by the current clef! Nonetheless, other commands such as `\Note` or `\Acc` are not affected and need absolute height values.



```
\begin{mousik}
      \Clef[t=4]
      \Key{1}
      \TimeSig{C|}{}

      \Acc{0}{8}
      \Note{1}{8}
      \Bar[|.]
\end{mousik}
```

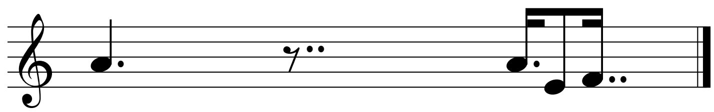
3.7 Other symbols

3.7.1 `\Dot`

The `\Dot` command prints any amount of dots in front of a note or rest. It accepts an optional argument and a compulsory argument.

The default number of dots is 1, but, as an optional argument, ** will produce 2 dots. The second, compulsory argument is the height of the dot.

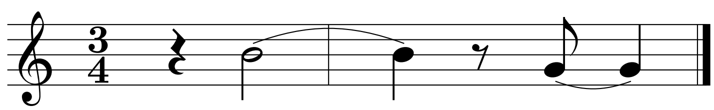
When there are several notes, the `\Dots` command might be useful. It accepts as an argument a list of pairs: symbols/height, where symbols might be * or **. Where there are no dots, the spot shall be left empty.



`\Dot{5}`	`\Dot[**]{7}`	`\Dots{*/5, , **/3}`
`\Note{4}{5}`	`\Rest{8}`	`\Note{|=,-,=|}{5,2,3}`

3.7.2 `\Tie`

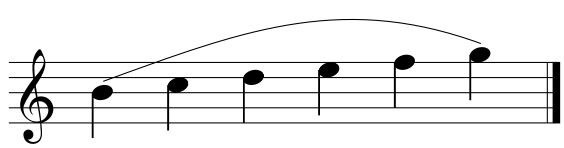
The `\Tie` command draws a tie between two notes. It has a compulsory argument, the height of both notes, and an optional one: the separation between the two notes. The default value is 1.



`\Tie[3]{6}` `\Tie{4}`

3.7.3 The `Slurred` environment

The `Slurred` environment draws a slur between the notes placed inside. It has two arguments: the heights of both beginning and end notes.

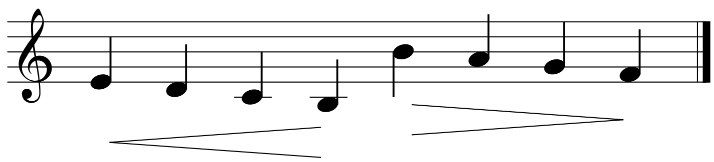


Fix angle

3.7.4 The `Crescendo` and `Decrescendo` environments

The `Crescendo` and `Decrescendo` environments draws a crescendo or decrescendo symbols, respectively, between the notes placed inside. It has an optional argument: the height of the symbol. The default value is 0.

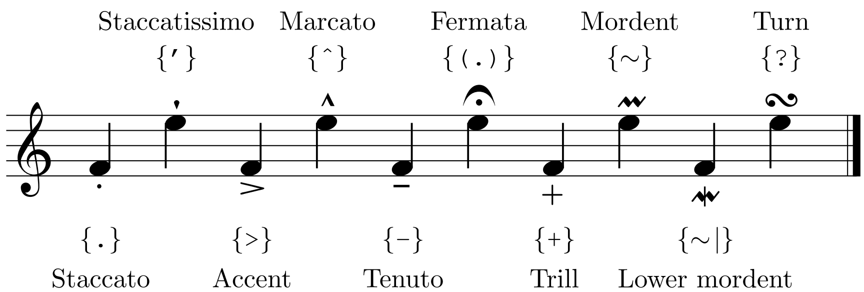
Half line

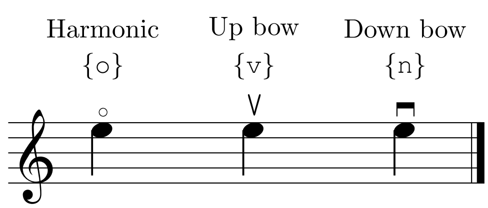


3.7.5 `\Artic` and `\Symbol`

The `\Artic` command prints an articulation symbol on a successive note. It accepts two arguments. The first one is the type of articulation.

Articsss

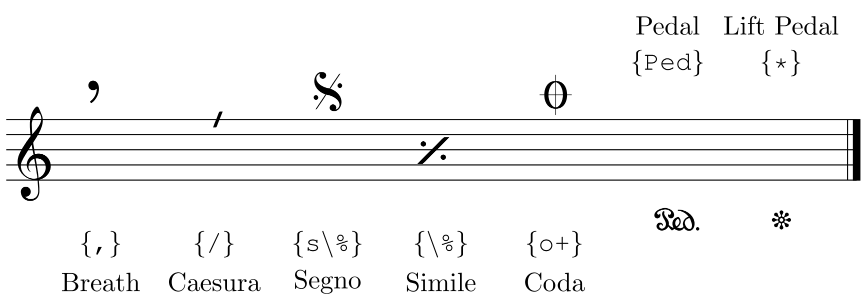




The second one is the height of the symbol, which might or might not be the height of the corresponding note.

When there are several notes, the `\Artics` command might be useful. It accepts as an argument a list of pairs: symbols/height. Where there are no symbols, the spot shall be left empty.

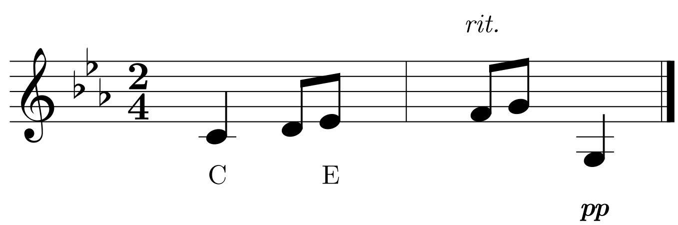
The `\Symbol` command is similar to the `\Artic` command, but it only accepts the type of symbol and not the height. These are symbols that have a fixed position on the score.



3.8 \Text and \Dynamics

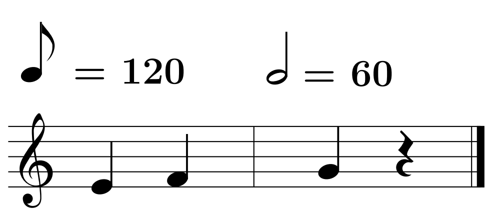
The `\Text` command allows the printing of any text string. It has a compulsory argument, but also an optional one where the user may choose the height of the string: `\Text[2]{\it rit.}`.

To write anything in the usual font of dynamics, the `\Dynamics` command will work the same: `\Dynamics[-0.5]{pp}`.



3.9 \Tempo

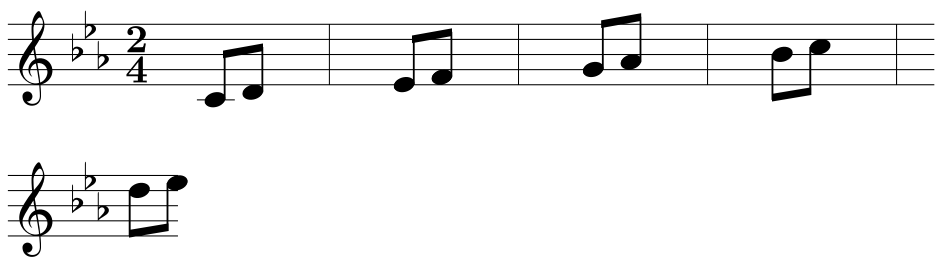
The `\Tempo` command prints a metronome tempo indicator. It has two arguments: the duration of the note and the numerical tempo which will be printed.



4 Tips and observations

4.1 Overflow of lines

The mousik package is thought to be used for small fragments of music. The environment has a maximum length allowed, and whenever it is exceeded an overflow happens. The line is broken at that exact point and, if it does not coincide with a bar, something like this may happen:



The best way to avoid the error is by introducing extra spaces manually. The overflow will also automatically print the previous clef and key.

4.2 How do spaces work?

Spaces serve as the manual progression of an invisible cursor during the printing. A negative

explain spaces

4.3 What if it becomes too cumbersome: your own commands

If a concrete rhythm is constantly used on a music piece.

explain

4.4 On your own: working with the **tikz** package

explain tikz

5 A very long and complete example

choose example


= 62
5
5
6
pp
3
f